QQ-T-351C
October 3, 1974
SUPERSEDING
Fed. Spec. QQ-T-351B
June 14, 1963

FEDERAL SPECIFICATION

TIN; PHOSPHOR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one composition of phosphor tin.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

QQ-T-371 - Tin, Pig.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.

FSC 9650

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Office in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of phosphor tin shall be as follows: Phosphorus - 3.5 percent minimum, phosphorus plus tin - 99.5 percent minimum.

3.2 Material. The tin used in the manufacture of phosphor tin shall conform to QQ-T-371 and shall be of uniform quality throughout.

3.3 Form. Phosphor tin shall be furnished in the form of notched slabs or ingots of such size that they can be conveniently handled.

3.4 Marking. Each slab or ingot shall be legibly and indelibly marked or stamped with the following information, when practical:

QQ-T-351.
Manufacturer's name and trademark.

3.5 Workmanship. The material shall be clean and free from defects injurious to the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the

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specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all material from one melt, but not more than 1000 pounds.

4.3 Sampling.

4.3.1 For chemical analysis. A sample for chemical analysis shall be taken from each lot. It shall consist of not less than five ounces of drillings taken from sound metal below the surface and shall be fine and clean, free from oil, grit, and other foreign matter.

4.3.2 For visual examination. From each lot of material, samples shall be selected for visual examination in accordance with MIL-STD-105 at inspection level I. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.4 Examination.

4.4.1 Visual. Each sample selected in accordance with 4.3.2 shall be subjected to surface examination for workmanship requirements.

4.4.2 Preparation for shipment. Examination of the packaging, packing, and marking for shipment shall be made for conformance with the requirements of section 5.

4.5 Test methods.

4.5.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or 112 of Fed. Test Method Std. No. 151. A single analysis of a composite sample may be made. In case of dispute, referee analysis shall be by method 111.

4.6 Rejection.

4.6.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.2 for that sample size, the entire lot shall be rejected subject to the provisions of the section on "Acceptance and Rejection" of MIL-STD-105.

4.6.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5, subject to the provisions of the section on "Rejection and Retests" of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

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5.1 Packaging and packing, level C. The products shall be packaged and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.2 Marking (see 6.2 and 6.3).

5.2.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Phosphor tin is used as an alloying additive in the melting and preparation of copper alloys.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Special marking, when required (see 5.2).

6.3 The requirements for product identification marking (see 3.4) and for packing and marking for shipment (see 5.1 and 5.2) specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

MILITARY CUSTODIANS:
Army - None
Navy - None
Air Force - None

Review activities:
Army - None
Navy - None
Air Force - None

User activities:
Army - None
Navy - None
Air Force - None

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing activity:
Army - MR

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 25 cents each.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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